

Installing Oracle Linux 7

Objectives

After completing this lesson, you should be able to:

- Obtain the Oracle Linux operating system software
- Describe the Anaconda installer
- Install Oracle Linux
- Complete initial setup
- Upgrade from Oracle Linux 6
- Describe steps to launch an Oracle Cloud Infrastructure instance



Obtaining Oracle Linux

- Obtain Oracle Linux from:
 - Oracle Software Delivery Cloud: <https://edelivery.oracle.com/linux/>
 - Oracle Linux yum server: <http://yum.oracle.com/>
 - Source and debug information available from: <http://oss.oracle.com/>
- Obtain errata for free from <http://yum.oracle.com/>.
 - Subscribe to the Oracle Linux errata mailing list from this site.
- Oracle Linux 7 is available for x86 64-bit systems.

Oracle Software Delivery Cloud

Oracle Linux download page: Example

You may download files:

- Using the download manager - Select the checkboxes next to the desired files, then click 'Download'
- Individually - Click the file name to download

[Back](#)[Print](#)[View Digest Details](#)[WGET Options](#)[Restore](#)[Download](#)

☐	Oracle Linux 7.5.0.0.0	
☐	Oracle Linux 7.5.0.0.0 for x86 64 bit	
☒	V975332-01.zip Readme for Driver Update Disk, 1.1 KB	
☐	V975333-01.iso Driver Update Disk for Oracle Linux 7 x86_64, 3.8 MB	
☐	V975334-01.zip Oracle Container Services for use with Kubernetes 1.1.9.1, 633.2 MB	
☐	V975335-01.iso Oracle VirtIO Drivers Version for Microsoft Windows 1.1.2, 59.2 MB	
☐	V975336-01.zip Oracle Container Services for use with Kubernetes 1.1.8, 713.1 MB	
☐	V975363-01.iso Oracle Linux Release 7 Update 5 Boot ISO image for x86 (64 bit), 540.0 MB	
☐	V975364-01.iso Oracle Linux Release 7 Update 5 UEK Boot ISO image for x86 (64 bit), 566.0 MB	
☐	V975365-01.iso Oracle Linux Release 7 Update 5 source DVD 1, 3.4 GB	
☐	V975366-01.iso Oracle Linux Release 7 Update 5 source DVD 2, 4.1 GB	
☒	V975367-01.iso Oracle Linux Release 7 Update 5 for x86 (64 bit), 4.1 GB	

Total 10 distinct files Total Distinct File Size 14.1 GB

NOTE: Some downloaded parts may be split into more than one file.

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Anaconda Installer

Anaconda:

- Is the installation program used by Oracle Linux
- Runs in textual or graphical mode
- Supports installation from local or remote sources
 - CD, DVD, USB drive, or images stored on a hard drive
 - NFS, HTTP, or FTP

The installation can be automated with Kickstart for unattended installation.

Oracle Linux Installation Menu



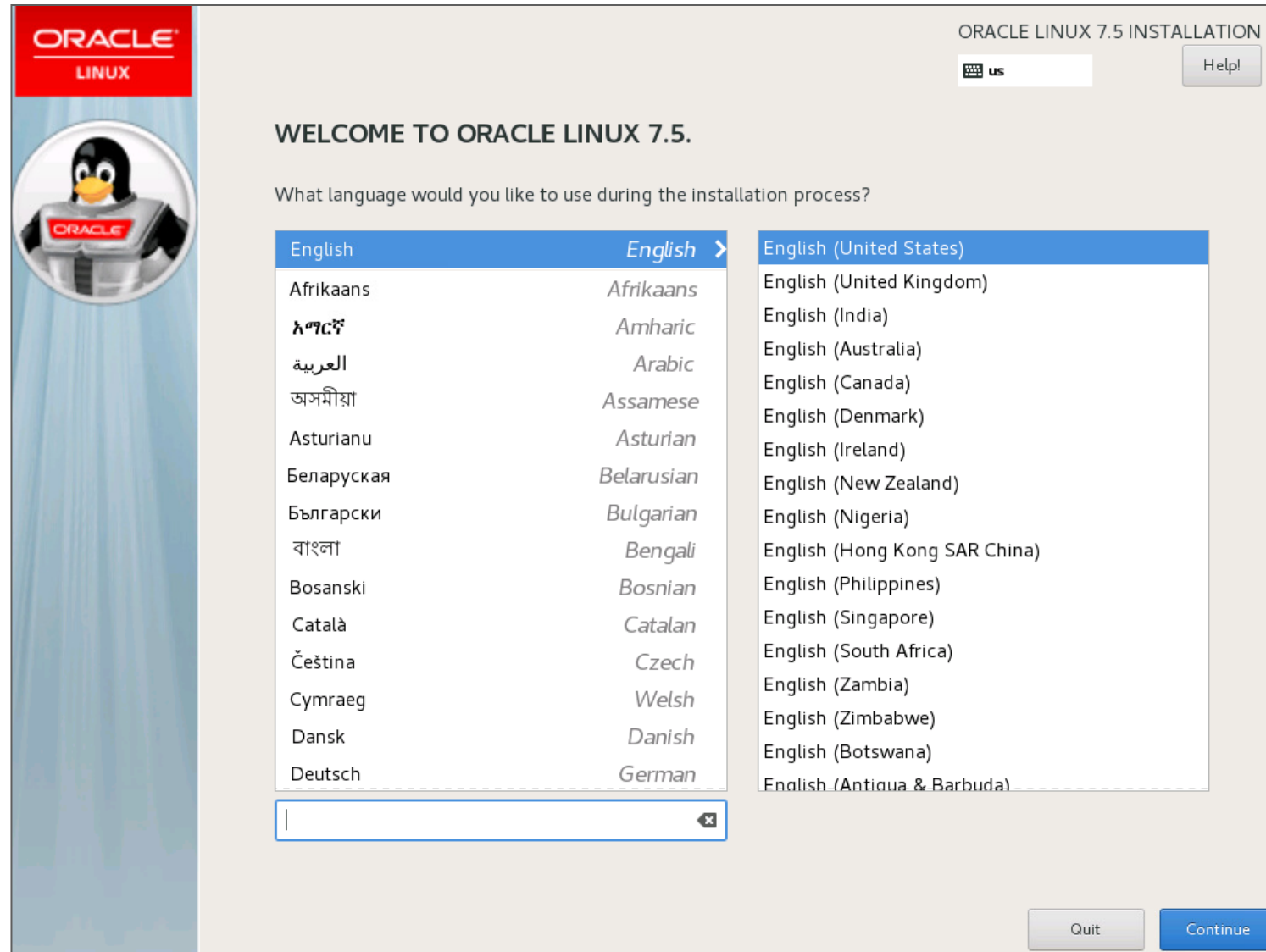
Boot Options

- Press Esc at the installation menu to display the `boot :` prompt.
- Use the following syntax to provide boot options:

```
boot: linux option1 option2 option3
```

- Some examples of boot settings include:
 - Language
 - Display resolution
 - Interface type
 - Installation method
 - Network settings

Welcome to Oracle Linux



Installation Summary



INSTALLATION SUMMARY

ORACLE LINUX 7.5 INSTALLATION

us

Help!

LOCALIZATION



DATE & TIME
Americas/New York timezone



KEYBOARD
English (US)



LANGUAGE SUPPORT
English (United States)

SOFTWARE



INSTALLATION SOURCE
Local media



SOFTWARE SELECTION
Minimal Install

SYSTEM



INSTALLATION DESTINATION
No disks selected



NETWORK & HOST NAME
Not connected



KDUMP
Kdump is enabled



SECURITY POLICY
No profile selected

Quit

Begin Installation

 Please complete items marked with this icon before continuing to the next step.

Date and Time Configuration

DATE & TIME

Done

ORACLE LINUX 7.5 INSTALLATION

us

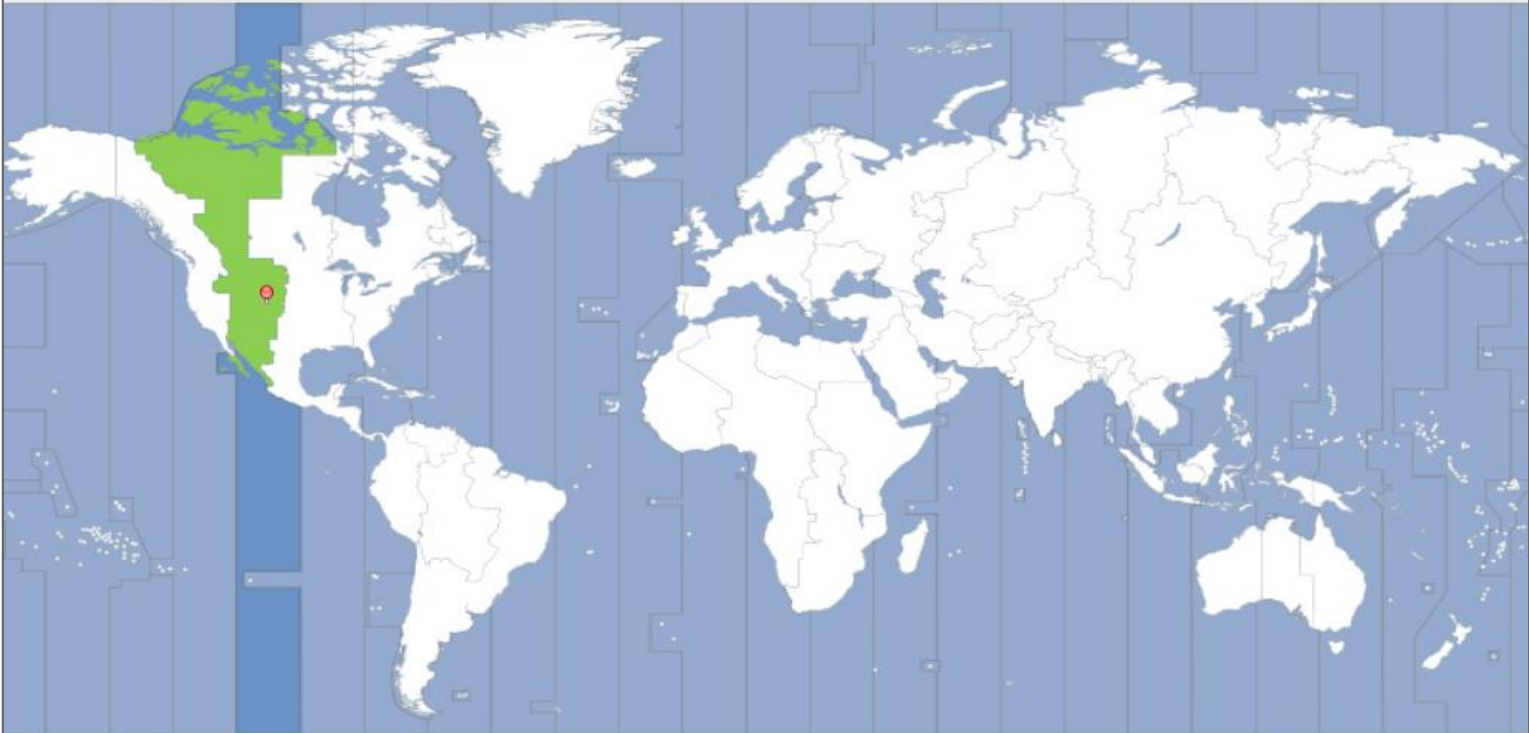
Help!

Region: Americas

City: Denver

Network Time

OFF



09:48 PM

24-hour

AM/PM

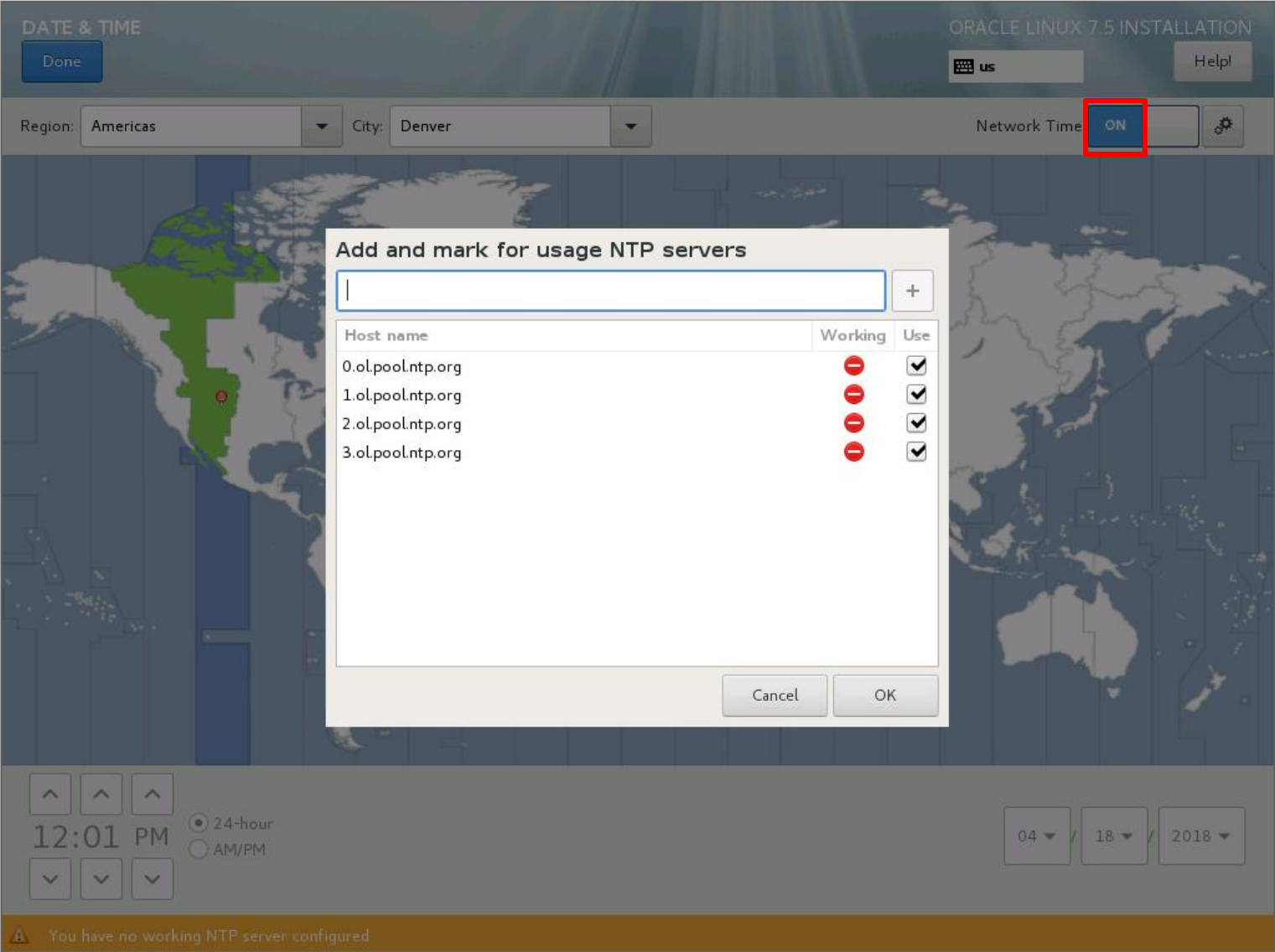
04

18

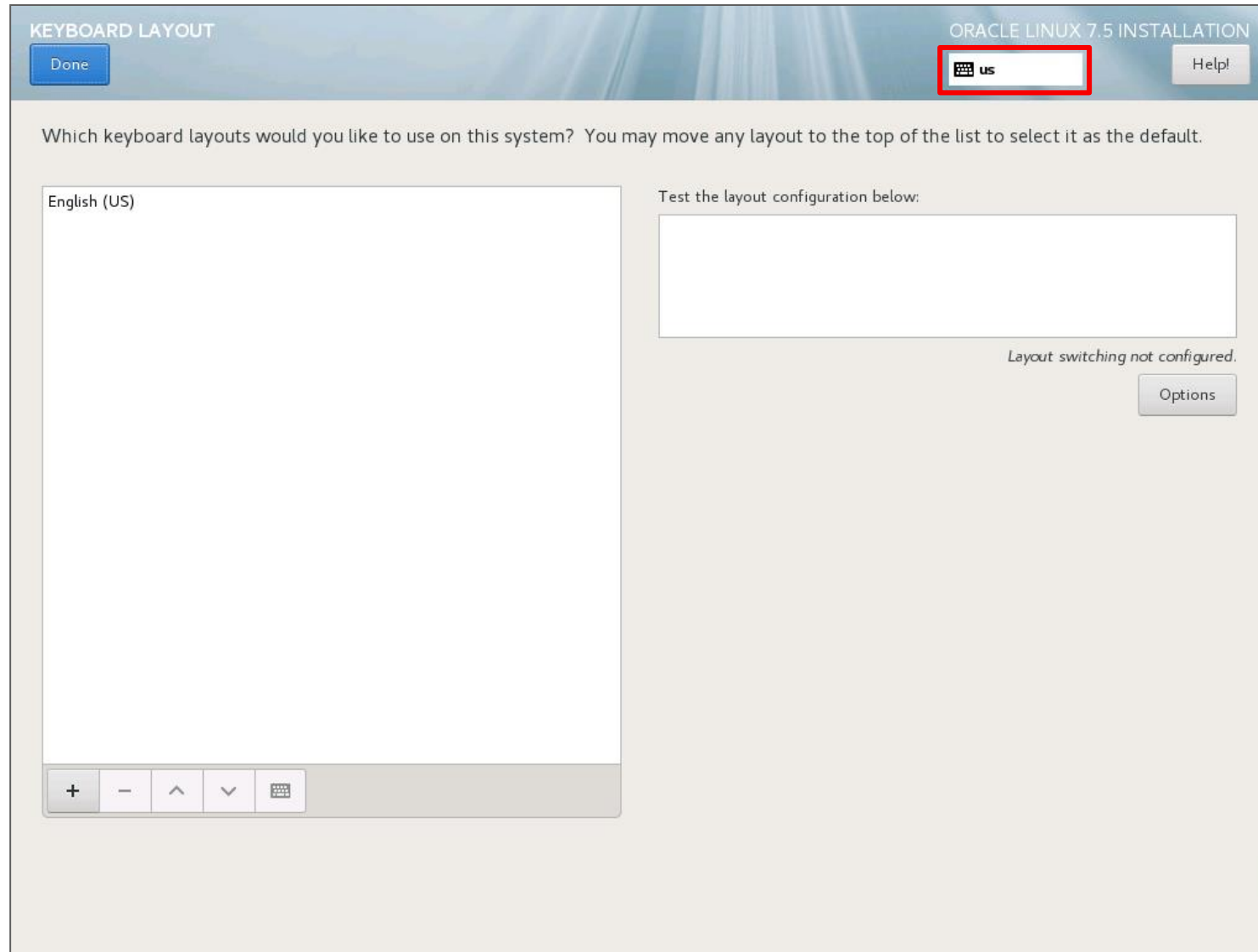
2018

You need to set up networking first if you want to use NTP

NTP Configuration



Keyboard Layout



The image shows a window titled "KEYBOARD LAYOUT" from the "ORACLE LINUX 7.5 INSTALLATION" process. The window has a blue header bar with a "Done" button on the left and a "Help!" button on the right. In the center of the header bar, there is a small box containing a keyboard icon and the text "us", which is highlighted with a red rectangle. Below the header bar, the main content area has a light beige background. It starts with the text: "Which keyboard layouts would you like to use on this system? You may move any layout to the top of the list to select it as the default." Below this text is a large, empty rectangular box. To the right of this box, the text "Test the layout configuration below:" is followed by another empty rectangular box. Below the right box, the text "Layout switching not configured." is displayed. At the bottom right of the main content area is an "Options" button. At the bottom left of the large empty box, there is a small toolbar with five buttons: a plus sign (+), a minus sign (-), an up arrow (^), a down arrow (v), and a keyboard icon.

Language Support

LANGUAGE SUPPORT

ORACLE LINUX 7.5 INSTALLATION

Done

 **us**

Help

Select additional language support to be installed:

☒ English	English	☒ English (United States)
☐ Español	Spanish	☐ English (United Kingdom)
☐ Eesti	Estonian	☐ English (India)
☐ Euskara	Basque	☐ English (Australia)
☐ فارسی	Persian	☐ English (Canada)
☐ Suomi	Finnish	☐ English (Denmark)
☐ Français	French	☐ English (Ireland)
☐ Galego	Galician	☐ English (New Zealand)
☐ ગુજરાતી	Gujarati	☐ English (Nigeria)
☐ हिन्दी	Hindi	☐ English (Hong Kong SAR China)
☐ Hrvatski	Croatian	☐ English (Philippines)
☐ Magyar	Hungarian	☐ English (Singapore)
☐ Interlingua	Interlingua	☐ English (South Africa)
☐ Bahasa Indonesia	Indonesian	☐ English (Zambia)
☐ Íslenska	Icelandic	☐ English (Zimbabwe)
☐ Italiano	Italian	☐ English (Botswana)
☐ 日本語	Japanese	☐ English (Antigua & Barbuda)
☐ ქართული	Georgian	

Software Installation Source

INSTALLATION SOURCE

ORACLE LINUX 7.5 INSTALLATION

Done

us

Help!

Which installation source would you like to use?

☒ Auto-detected installation media:

Device: sr0
Label: OL-7.5_Server.x86_64

Verify

☐ On the network:

http://

Proxy setup...

☐ This URL refers to a mirror list.

Additional repositories

Enabled	Name
☒	Server-HighAvailability
☐	Server-ResilientStorage

+

-

↺

Name:

Server-HighAvailability

file://

/run/install/repo/addons/HighAvailability

☐ This URL refers to a mirror list.

Proxy URL:

User name:

Password:

You need to configure the network to use a network installation source.

Selecting the Software to Install

SOFTWARE SELECTION

Done

ORACLE LINUX 7.5 INSTALLATION

us

Help!

Base Environment

☐ Minimal Install
Basic functionality.

☐ Infrastructure Server
Server for operating network infrastructure services.

☐ File and Print Server
File, print, and storage server for enterprises.

☐ Basic Web Server
Server for serving static and dynamic internet content.

☐ Virtualization Host
Minimal virtualization host.

☒ Server with GUI
Server for operating network infrastructure services, with a GUI.

Add-Ons for Selected Environment

☒ FTP Server
Allows the system to act as an FTP server.

☒ File and Storage Server
CIFS, SMB, NFS, iSCSI, iSER, and iSNS network storage server.

☐ Hardware Monitoring Utilities
A set of tools to monitor server hardware.

☐ Identity Management Server
Centralized management of users, servers and authentication policies.

☐ Infiniband Support
Software designed for supporting clustering and grid connectivity using RDMA-based InfiniBand and iWARP fabrics.

☐ Java Platform
Java support for the Oracle Linux Server and Desktop Platforms.

☐ KDE
The KDE Plasma Workspaces, a highly-configurable graphical user interface which includes a panel, desktop, system icons and desktop widgets, and many powerful KDE applications.

☐ Large Systems Performance
Performance support tools for large systems.

☐ Load Balancer
Load balancing support for network traffic.

☐ Mainframe Access
Tools for accessing mainframe computing resources.

☐ MariaDB Database Server
The MariaDB SQL database server, and associated packages.

☒ Network File System Client
Enables the system to attach to network storage.

☐ Performance Tools
Tools for diagnosing system and application-level performance problems.

☐ PostgreSQL Database Server
The PostgreSQL SQL database server, and associated packages.

Installation Destination

INSTALLATION DESTINATION

ORACLE LINUX 7.5 INSTALLATION

Done

us

Help!

Device Selection

Select the device(s) you'd like to install to. They will be left untouched until you click on the main menu's "Begin Installation" button.

Local Standard Disks

16 GiB

xvda / 16 GiB free

5120 MiB

xvdb / 5120 MiB free

5120 MiB

xvdd / 5120 MiB free

Disks left unselected here will not be touched.

Specialized & Network Disks

Add a disk...

Disks left unselected here will not be touched.

Other Storage Options

Partitioning

☒ Automatically configure partitioning.

☐ I will configure partitioning.

☐ I would like to make additional space available.

Encryption

☐ Encrypt my data. You'll set a passphrase next.

[Full disk summary and boot loader...](#)

1 disk selected; 16 GiB capacity; 16 GiB free [Refresh...](#)

Automatic Partitioning

Done

us

Help!

New Oracle Linux 7.5 Installation

SYSTEM

/boot	1024 MiB	>
xvda1		
/	13.39 GiB	
ol-root		
swap	1638.4 MiB	
ol-swap		

+ - ↺

AVAILABLE SPACE

992.5 KiB

TOTAL SPACE

16 GiB

[1 storage device selected](#)

xvda1

Mount Point:

/boot

Device(s):

(xvda)

Desired Capacity:

1024 MiB

Modify...

Device Type:

Standard Partition

☐ Encrypt

File System:

xf

☒ Reformat

Label:

Name:

xvda1

Update Settings

Note: The settings you make on this screen will not be applied until you click on the main menu's 'Begin Installation' button.

Reset All

Manual Partitioning

MANUAL PARTITIONING

ORACLE LINUX 7.5 INSTALLATION

Done

us

Help!

▼ New Oracle Linux 7.5 Installation

You haven't created any mount points for your Oracle Linux 7.5 installation yet. You can:

- [Click here to create them automatically.](#)
- Create new mount points by clicking the '+' button.

New mount points will use the following partitioning scheme:

LVM

+

-

↺

AVAILABLE SPACE

16 GiB

TOTAL SPACE

16 GiB

[1 storage device selected](#)

When you create mount points for your Oracle Linux 7.5 installation, you'll be able to view their details here.

Reset All

Summary of Partitioning

SUMMARY OF CHANGES				
Your customizations will result in the following changes taking effect after you return to the main menu and begin installation:				
Order	Action	Type	Device Name	Mount point
1	Destroy Format	Unknown	xvda	
2	Create Format	partition table (MSDOS)	xvda	
3	Create Device	partition	xvda1	
4	Create Device	partition	xvda2	
5	Create Device	partition	xvda3	
6	Create Device	partition	xvda5	
7	Create Format	swap	xvda5	
8	Create Format	ext4	xvda3	/home
9	Create Format	ext4	xvda2	/
10	Create Format	ext4	xvda1	/boot

Cancel & Return to Custom PartitioningAccept Changes

Network and Host Name Configuration

NETWORK & HOST NAME

ORACLE LINUX 7.5 INSTALLATION

Done us Help!

Ethernet (eth0)

Ethernet (eth1)

+ -

Ethernet (eth0)
Disconnected

Hardware Address 00:16:3E:00:01:03

Speed

Subnet Mask 127.0.0.1

☐ OFF

Configure...

Host name: Apply

Current host name: localhost

Network Connection Settings

Editing eth0

Connection name:

General Ethernet 802.1X Security DCB Proxy **IPv4 Settings** IPv6 Settings

Method:

Addresses

Address	Netmask	Gateway
192.0.2.103	24	192.0.2.1

DNS servers:

Search domains:

DHCP client ID:

☐ Require IPv4 addressing for this connection to complete

Security Policy

SECURITY POLICY

ORACLE LINUX 7.5 INSTALLATION

Done

us

Help!

Change content

Apply security policy: **ON**

Choose profile below:

Standard System Security Profile
This profile contains rules to ensure standard security baseline of Oracle Linux 7 system. Regardless of your system's workload all of these checks should pass.

PCI-DSS v3 Control Baseline for Oracle Linux 7
This is a *draft* profile for PCI-DSS v3.

Oracle Profile for Cloud Providers
This is a *draft* SCAP profile for Cloud Providers.

Common Profile for General-Purpose Systems
This profile contains items common to general-purpose desktop and server installations.

DISA STIG for Oracle Linux 7
This is a *draft* profile for STIG for Oracle Linux 7.

United States Government Configuration Baseline (USGCB / STIG) - DRAFT
This profile is developed in partnership with the U.S. National Institute of Standards and Technology (NIST), U.S. Department of Defense, the National Security Agency, and Red Hat. The USGCB is intended to be the core set of security related configuration settings by which all federal agencies should comply.

This baseline implements configuration requirements from the following documents:

Select profile

Changes that were done or need to be done:

💡 No profile selected

KDUMP

KDUMP

Done

ORACLE LINUX 7.5 INSTALLATION

us

Help!

Kdump is a kernel crash dumping mechanism. In the event of a system crash, kdump will capture information from your system that can be invaluable in determining the cause of the crash. Note that kdump does require reserving a portion of system memory that will be unavailable for other uses.

☒ Enable kdump

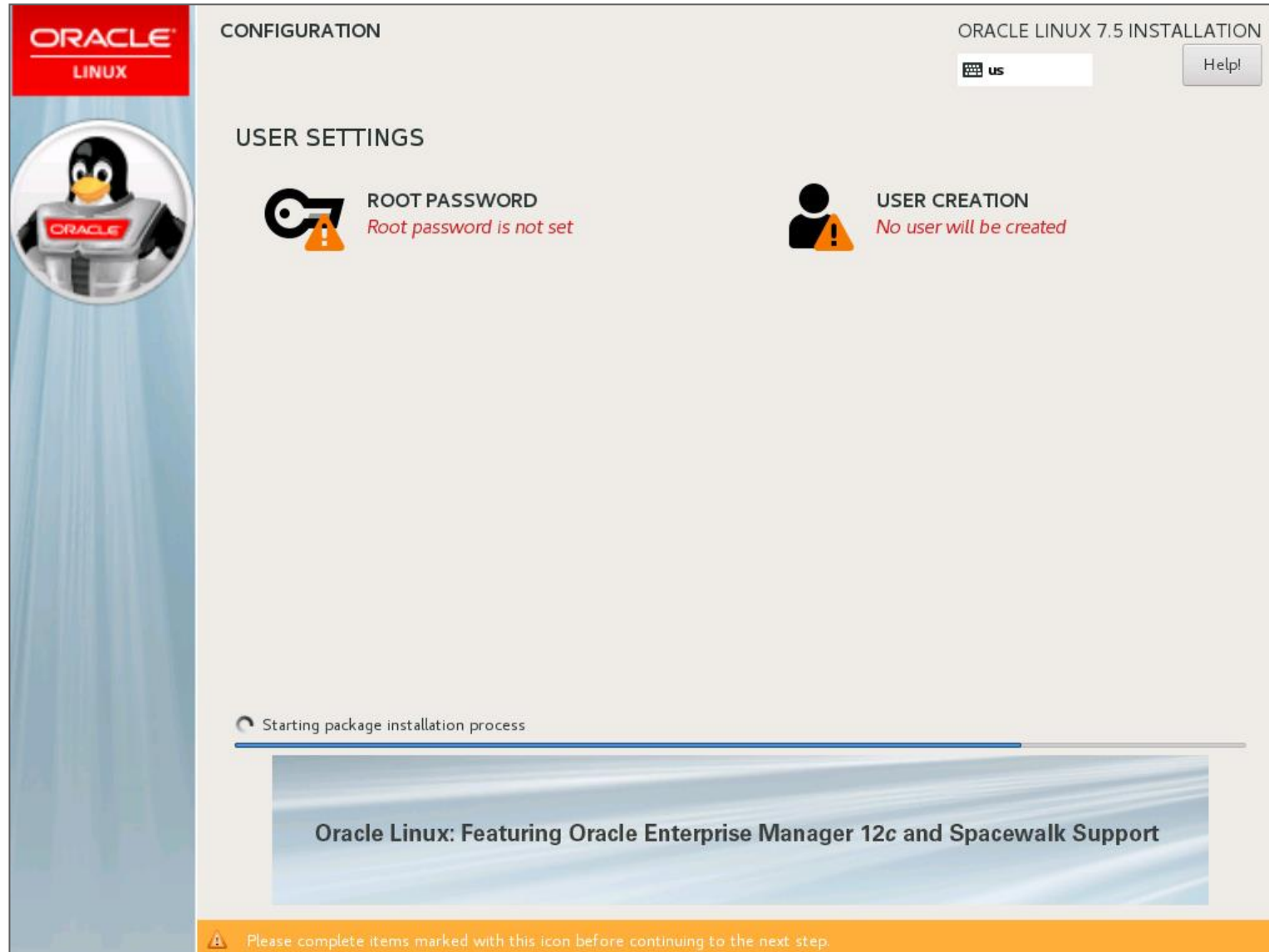
Kdump Memory Reservation: ☒ Automatic ☐ Manual

Memory To Be Reserved (MB):

Total System Memory (MB): 1998
Usable System Memory (MB): 1870

- Crash dumping mechanism for the Linux kernel
- Enabled by default

Completing the Installation



Setting the root Password

ROOT PASSWORD ORACLE LINUX 7.5 INSTALLATION

[Done](#) us [Help!](#)

The root account is used for administering the system. Enter a password for the root user.

Root Password:

Weak

Confirm:

 The password you have provided is weak. The password fails the dictionary check - it is based on a dictionary word. You will have to press Done twice to confirm it.

Creating an Initial User

CREATE USER ORACLE LINUX 7.5 INSTALLATION

Done us Help!

Full name

User name

Tip: Keep your user name shorter than 32 characters and do not use spaces.

☐ Make this user administrator

☒ Require a password to use this account

Password

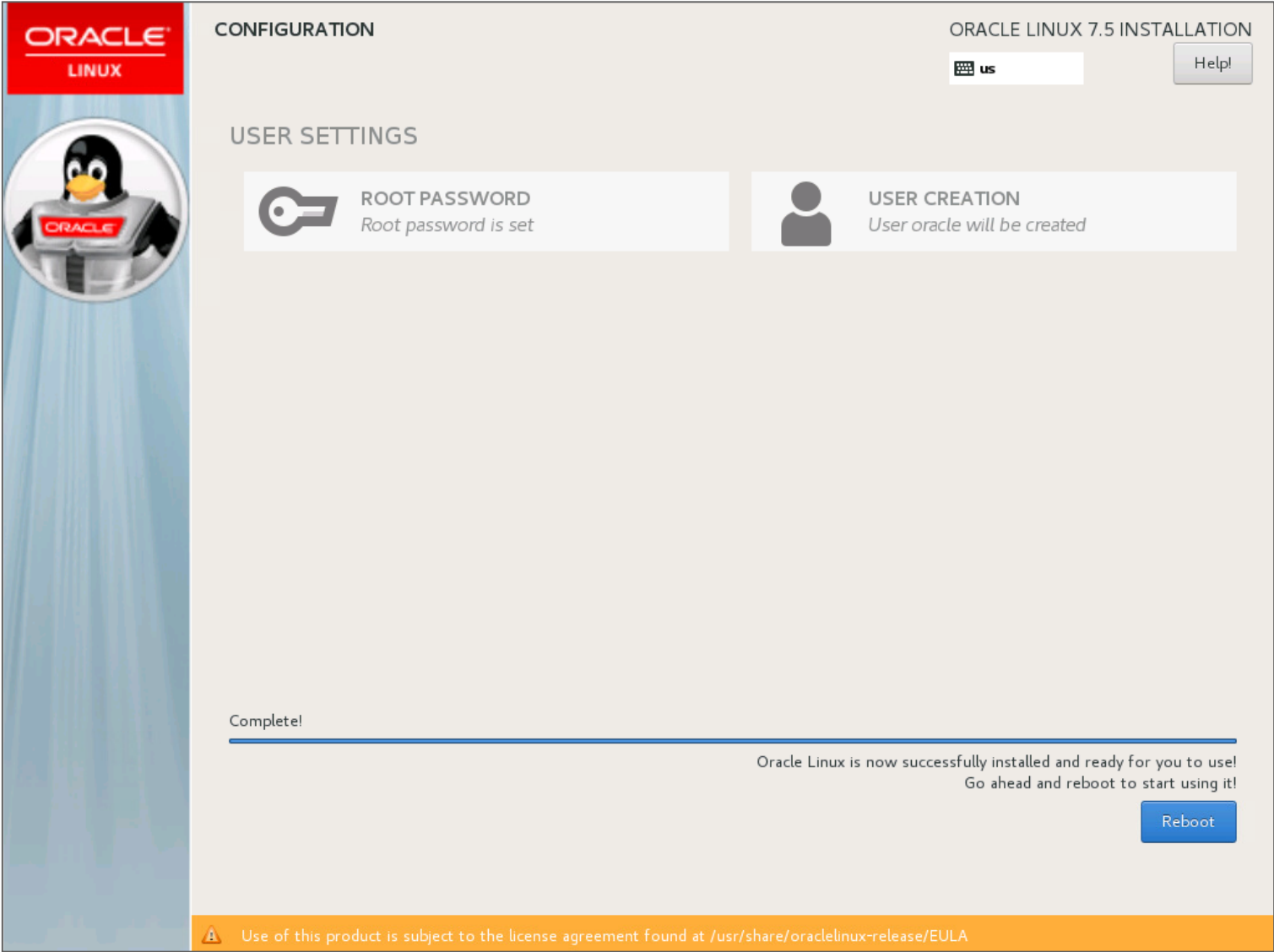
Weak

Confirm password

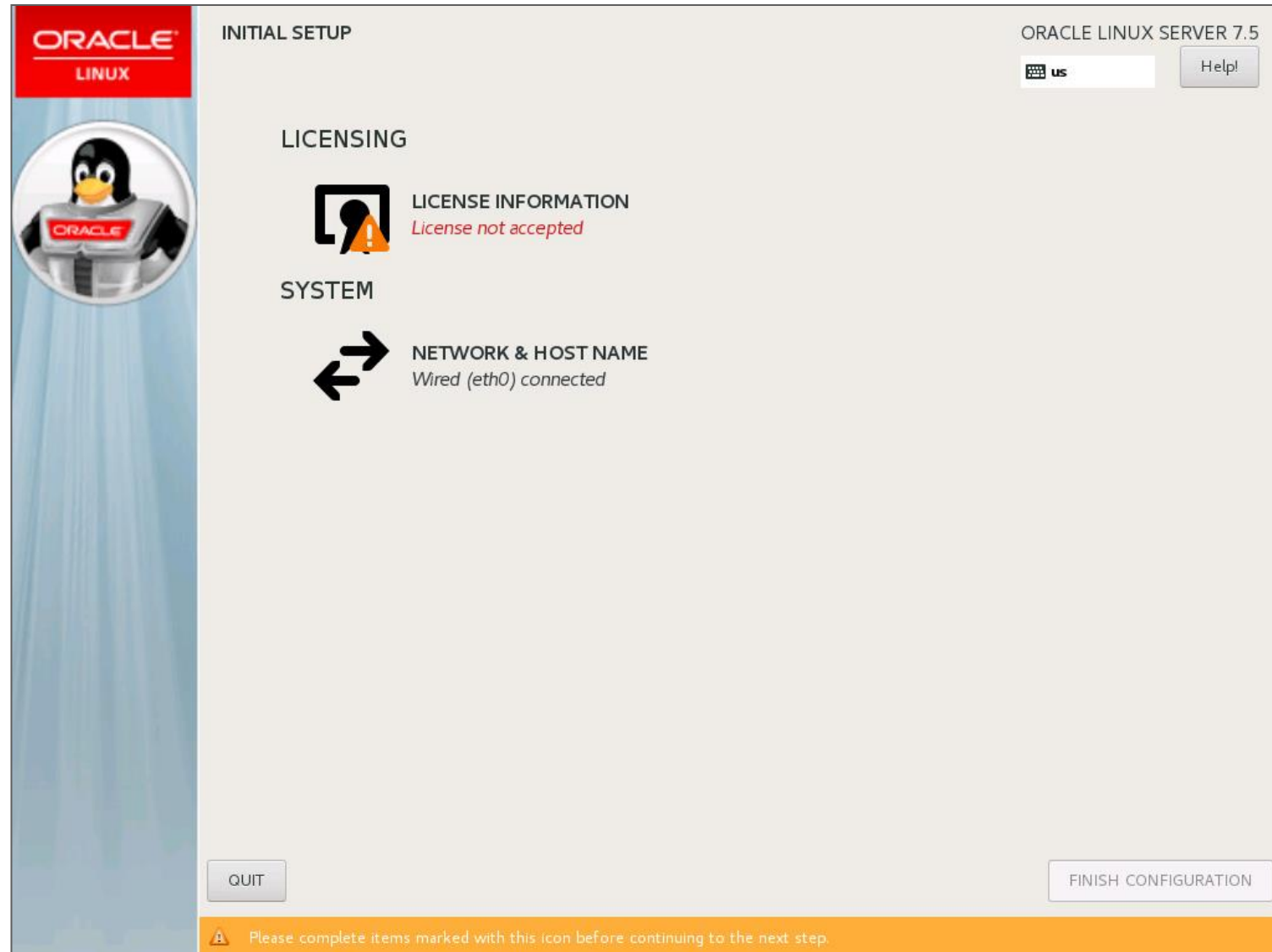
Advanced...

 The password you have provided is weak: The password contains the user name in some form You will have to press Done twice to confirm it..

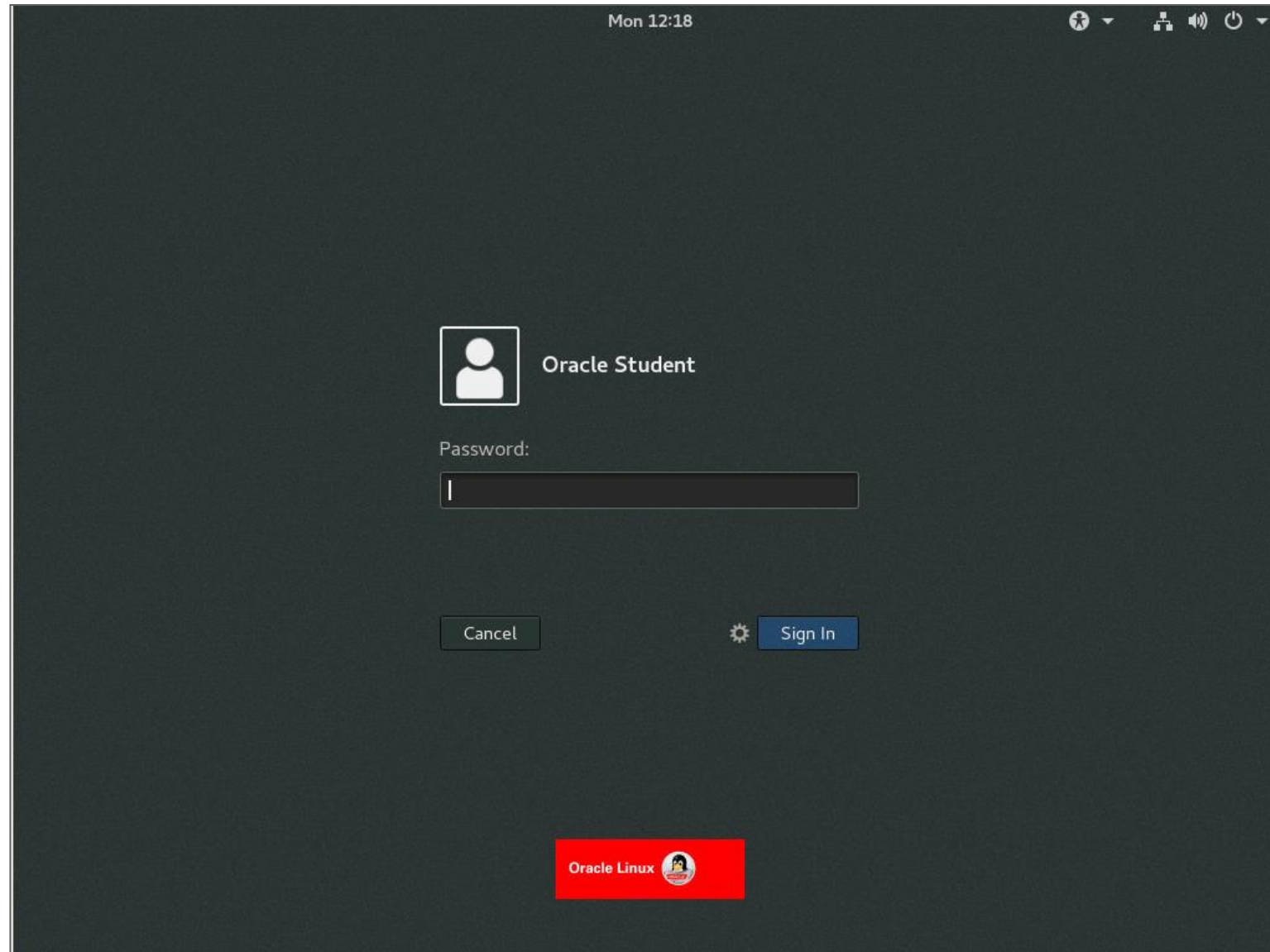
Installation Complete



Initial Setup



GUI Login Window



Inplace Upgrade from Oracle Linux 6

An inplace upgrade to Oracle Linux 7 from Oracle Linux 6 Update 5 or later is possible under the following conditions:

- The system meets the minimum installation requirements as described in the Oracle Linux 7 Release Notes.
- The Oracle Linux 6 system has been completely updated from the `ol6_x86_64_latest` channel or `ol6_latest` repository.
- UEK R3 or UEK R4 is installed on the system to be upgraded and is the default boot kernel.
 - Upgrading from UEK R2 is not supported.
- No Oracle product stack is present on the system.
- The system to be upgraded has the Minimal Install base environment.
- See “Upgrading an Oracle Linux System” in the *Oracle Linux 7 Installation Guide* for complete instructions.

Launching an Oracle Cloud Infrastructure Instance

- Launching an instance does not require system installation, as described in this lesson.
- An image and shape are selected, providing an operating system template as well as CPU and memory resources.
- A Virtual Cloud Network and subnet are selected.
- One or more SSH public keys are provided.
- The operating system with accompanying resources is deployed for you.
- After the operating system boots, you can log in to the instance.

Quiz



To enable NTP during installation, you must be connected to a network.

- a. True
- b. False



Quiz



The default File System type when partitioning disks is:

- a. ext4
- b. ext3
- c. xfs
- d. vfat



Summary

In this lesson, you should have learned how to:

- Obtain the Oracle Linux operating system software
- Describe the Anaconda installer
- Install Oracle Linux
- Complete initial setup
- Upgrade from Oracle Linux 6
- Describe steps to launch an Oracle Cloud Infrastructure instance



Practice 4: Overview

This practice covers the following topics:

- Installing Oracle Linux
- Completing initial setup
- Booting up the host03 VM guest